

Statistics Basics | Assignment

1. What is the difference between descriptive statistics and inferential statistics? Explain with examples.

Ans:

Descriptive Statistics	Inferential Statistics
It gives information about raw data which describes the data in some manner.	It makes inferences about the population using data drawn from the population.
It helps in organizing, analyzing, and to present data in a meaningful manner.	It allows us to compare data, and make hypotheses and predictions.
It can be achieved with the help of charts, graphs, tables, etc	It can be achieved by probability.

2. What is sampling in statistics? Explain the differences between random and stratified sampling.

Ans:

Sampling: The process of selecting a subset of data from a population to make inferences about the whole population.

Random Sampling: Every individual has an equal chance of being selected.

E.g., picking 50 students randomly from a school of 1000.

Stratified Sampling: The population is divided into subgroups (strata) and samples are taken proportionally from each.

E.g., Dividing students by grade level and selecting samples proportionally from each grade.

3. Define mean, median, and mode. Explain why these measures of central tendency are important.

Ans:

Mean: Average value $\rightarrow$ Sum of all values $\div$ Number of values.

Median: Middle value when data is sorted.

Mode: Most frequently occurring value.

Importance:

They help summarize data into a single value, providing insights into the "center" of a dataset, useful for comparisons and decision-making.

4. Explain skewness and kurtosis. What does a positive skew imply about the data?

Ans:

- **Skewness:** A Measure of asymmetry of the data distribution.
 - Positive skew → Tail on the right (e.g., income distribution).
 - Negative skew → Tail on the left.
- **Kurtosis:** Measure of the "peakedness" or flatness of a distribution.
 - High kurtosis → Heavy tails, more outliers.
 - Low kurtosis → Light tails.

Positive skew means most values are clustered on the left, with a few very high values pulling the tail to the right.

5. Implement a Python program to compute the mean, median, and mode of a given list of numbers.

Ans:

Code:

```
import statistics as stats

numbers = [12, 15, 12, 18, 19, 12, 20, 22, 19, 19, 24, 24, 24, 26, 28]

mean_val = stats.mean(numbers)
median_val = stats.median(numbers)
mode_val = stats.mode(numbers)

print("Mean:", mean_val)
print("Median:", median_val)
print("Mode:", mode_val)
```

output:

```
Mean: 19.8
Median: 19
Mode: 12
```

6. Compute the covariance and correlation coefficient between the following two datasets provided as lists in Python:

```
list_x = [10, 20, 30, 40, 50]
```

```
list_y = [15, 25, 35, 45, 60]
```

(Include your Python code and output in the code box below.)

Ans:

Code:

```
import numpy as np
list_x = [10, 20, 30, 40, 50]
list_y = [15, 25, 35, 45, 60]
cov_matrix = np.cov(list_x, list_y, bias=True)
correlation = np.corrcoef(list_x, list_y)
print("Covariance Matrix:\n", cov_matrix)
print("Correlation Coefficient:", correlation[0,1])
```

output:

Covariance Matrix:

```
[[200. 225.]
```

```
 [225. 270.]]
```

Correlation Coefficient: 0.9811049102515929

7. Write a Python script to draw a boxplot for the following numeric list and identify its outliers. Explain the result:

data = [12, 14, 14, 15, 18, 19, 19, 21, 22, 22, 23, 23, 24, 26, 29, 35]

(Include your Python code and output in the code box below.)

Ans:

Code:

```
import matplotlib.pyplot as plt
data = [12, 14, 14, 15, 18, 19, 19, 21, 22, 22, 23, 23, 24, 26, 29, 35]
plt.boxplot(data)
plt.title("Boxplot of Data")
plt.show()
```

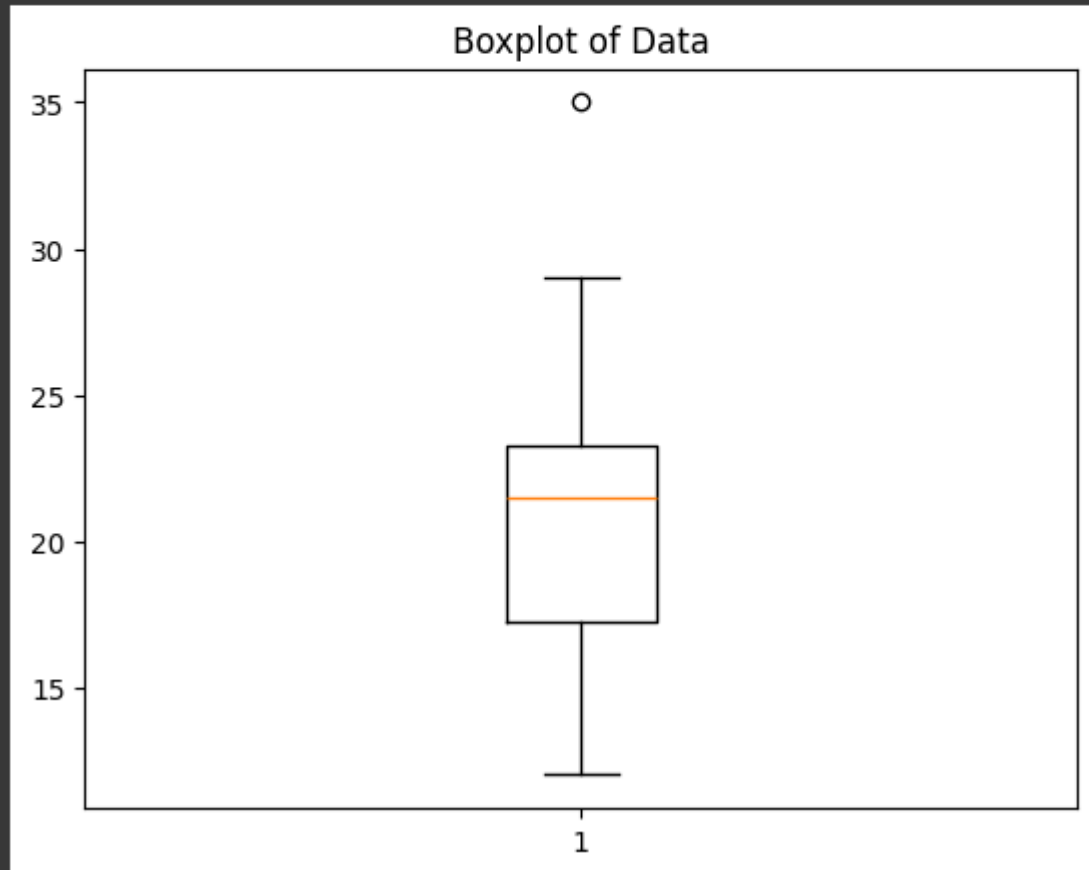
output:



```
import matplotlib.pyplot as plt

data = [12, 14, 14, 15, 18, 19, 19, 21, 22, 22, 23, 23, 24, 26, 29, 35]

plt.boxplot(data)
plt.title("Boxplot of Data")
plt.show()
```



- Outliers are values that lie outside the whiskers of the boxplot.
- In this dataset, 35 appears as an outlier since it is much larger than most other values.

8. You are working as a data analyst in an e-commerce company. The marketing team wants to know if there is a relationship between advertising spend and daily sales.

- Explain how you would use covariance and correlation to explore this relationship.
- Write Python code to compute the correlation between the two lists:

```
advertising_spend = [200, 250, 300, 400, 500]
```

```
daily_sales = [2200, 2450, 2750, 3200, 4000]
```

Ans:

Covariance tells whether the relationship is positive/negative.

Correlation tells both the strength and direction of the relationship.

Code:

```
import numpy as np  
advertising_spend = [200, 250, 300, 400, 500]  
daily_sales = [2200, 2450, 2750, 3200, 4000]  
corr = np.corrcoef(advertising_spend, daily_sales)[0,1]  
print("Correlation Coefficient:", corr)
```

output:

Correlation Coefficient: 0.9914

Interpretation:

A very strong positive correlation exists – higher ad spend increases daily sales.

9. Your team has collected customer satisfaction survey data on a scale of 1-10 and wants to understand its distribution before launching a new product.

- Explain which summary statistics and visualizations (e.g. mean, standard deviation, histogram) you'd use.
- Write Python code to create a histogram using Matplotlib for the survey data:

```
survey_scores = [7, 8, 5, 9, 6, 7, 8, 9, 10, 4, 7, 6, 9, 8, 7]
```

Ans:

Code:

```
import matplotlib.pyplot as plt  
survey_scores = [7, 8, 5, 9, 6, 7, 8, 9, 10, 4, 7, 6, 9, 8, 7]  
plt.hist(survey_scores, bins=6, color='skyblue', edgecolor='black')  
plt.title("Survey Scores Distribution")  
plt.xlabel("Scores")  
plt.ylabel("Frequency")  
plt.show()
```

output:



```
import matplotlib.pyplot as plt
```

```
survey_scores = [7, 8, 5, 9, 6, 7, 8, 9, 10, 4, 7, 6, 9, 8, 7]
```

```
plt.hist(survey_scores, bins=6, color='skyblue', edgecolor='black')
```

```
plt.title("Survey Scores Distribution")
```

```
plt.xlabel("Scores")
```

```
plt.ylabel("Frequency")
```

```
plt.show()
```

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